



Year 11 Mathematics Extension 1
Assessment Task 2
Term 3, Week 4, 2022

Name:

Teacher:

Thursday 11th August, 2022

Setter: NAI

General Instructions

- **Working time – 45 minutes plus 2 minutes reading time.**
- Answer all questions on the lined paper provided using black or blue pen.
- NESA approved calculators may be used.
- Show all relevant mathematical reasoning and/or calculations.
- **Total marks – 32.**
- Attempt Questions 1–6.

Question 1	(5 marks)	Marks
(a)	(i) Factorise $3a^2 - 10a + 3$.	1
	(ii) Hence or otherwise solve, $3(\log_3 x)^2 - 10\log_3 x + 3 = 0$, for x .	2
(b)	Differentiate $e^{3x}(x^2 + 1)$ with respect to x , leaving your answer fully factorised.	2

End of Question 1

Question 2	(5 marks) (Start a new page)	Marks
(a)	(i) Find the derivative of $y = e^x - x$.	1
	(ii) Hence find the equation of the tangent to the curve at the point where the curve cuts the y-axis.	2
(b)	Find the exact value of $\cos\left[\sin^{-1}\left(\frac{5}{13}\right) + \tan^{-1}\left(\frac{3}{5}\right)\right]$	2

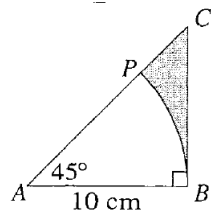
End of Question 2

Question 3 (5 marks) (Start a new page)**Marks**

(a) (i) Prove that $\frac{\sin 2x}{1 - \cos 2x} = \cot x$. 2

(ii) Hence find the value for $\cot 67\frac{1}{2}^\circ$ in simplest surd form. 1

- (b) In the diagram $\triangle ABC$ is right-angled at B , $AB = 10$ cm and $\angle A = 45^\circ$. The circular arc BP has a centre A and radius AB . It meets the hypotenuse AC at P .



(i) Find the exact area of sector ABP . 1

(ii) Hence find the exact area of the shaded portion BCP . 1

End of Question 3

Question 4 (5 marks) (Start a new page)**Marks**

Ice-cream is taken out of a freezer at -10°C into a room where the air temperature is 24°C . The rate at which the ice-cream warms follows Newton's law, that is $\frac{dT}{dt} = -k(T - 24)$, where k is a positive integer and t is measured in minutes and T in $^\circ\text{C}$.

(i) Show that the function $T = 24 - Ae^{-kt}$, where A is a constant, provides this rate of change. 1

(ii) Hence find the value of A . 1

(iii) The temperature of the ice-cream reaches 2°C in 1.5 hours. Find the exact value of k . 2

(iii) Find the temperature of the ice-cream after a further 1.5 hours correct to 2 decimal places. 1

End of Question 4

Question 5 (6 marks) (Start a new page)**Marks**

(a) (i) Prove that: $\frac{1}{\sqrt{x}} - \frac{1}{\sqrt{x+h}} = \frac{h}{(x+h)\sqrt{x} + x\sqrt{x+h}}$. **3**

(ii) Hence, find $\frac{d}{dx} \sqrt{\frac{2}{x}}$ from first principles. **3**

End of Question 5**Question 6** (6 marks) (Start a new page)**Marks**

(a) Solve $\sin x + \cos x = -1$, for $0 \leq x \leq 2\pi$, by using the substitution $t = \tan \frac{x}{2}$. **2**

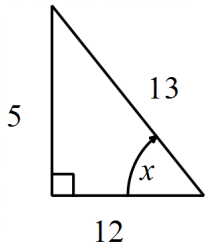
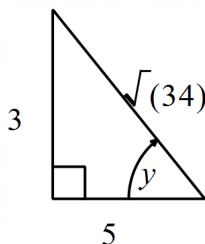
(b) Using t -formula, prove that: $\tan 2\theta(\sin 2\theta - \tan \theta) = \sin 2\theta \tan \theta$. **2**

(c) Using at least a third of a page, sketch the function $y = \sin(\cos^{-1} x)$. **2**

End of Assessment Task

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2022 Year 11 Mathematics Ext1 AT2 Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments
Question 1			
ai) $(3a-1)(a-3)$	1 mark	$\cos\left(\sin^{-1}\left(\frac{5}{13}\right) + \tan^{-1}\left(\frac{3}{5}\right)\right)$	
ii) $(3\log_3 x - 1)(\log_3 x - 3) = 0$ $3\log_3 x - 1 = 0$ or $\log_3 x - 3 = 0$ $\log_3 x = \frac{1}{3}$ or $\log_3 x = 3$ $x = 3^{\frac{1}{3}} = \sqrt[3]{3}$ or $x = 3^3 = 27$	1 mark 1 mark for both answers	$= \cos\left(\sin^{-1}\left(\frac{5}{13}\right)\right) \cos\left(\tan^{-1}\left(\frac{3}{5}\right)\right) - \sin\left(\sin^{-1}\left(\frac{5}{13}\right)\right) \sin\left(\tan^{-1}\left(\frac{3}{5}\right)\right)$ $= \frac{12}{13} \times \frac{5}{\sqrt{34}} - \frac{5}{13} \times \frac{3}{\sqrt{34}}$ $= \frac{45}{13\sqrt{34}} \times \frac{\sqrt{34}}{\sqrt{34}}$ $= \frac{45\sqrt{34}}{442}$	1 mark (no penalty for not rationalising)
b) let $y = e^{3x}(x^2 + 1)$ let $u = e^{3x}$ and $v = x^2 + 1$ $u' = 3e^{3x}$ and $v' = 2x$ $y' = 3e^{3x}(x^2 + 1) + e^{3x} \times 2x$ $y' = e^{3x}(3x^2 + 2x + 3)$	1 mark 1 mark	Alternate Solution Let $x = \sin^{-1}\left(\frac{5}{13}\right)$, $\sin x = \frac{5}{13}$ $y = \tan^{-1}\left(\frac{3}{5}\right)$, $\tan y = \frac{3}{5}$ $\cos(x + y) = \cos x \cos y - \sin x \sin y$ $= \frac{12}{13} \times \frac{5}{\sqrt{34}} - \frac{5}{13} \times \frac{3}{\sqrt{34}}$ $= \frac{45}{13\sqrt{34}} \times \frac{\sqrt{34}}{\sqrt{34}}$ $= \frac{45\sqrt{34}}{442}$	Use diagrams as starting point
Question 2			
2ai) $y' = e^x - 1$	1 mark		
aii) when $x = 0$, $y = e^0 - 0 = 1$ $\therefore$ the curve cuts the y -axis at $(0, 1)$ $\therefore$ the equation of the tangent is: $y - 1 = (e^0 - 1)(x - 0)$ $y = 1$	1 mark 1 mark		1 mark for the final answer
b)		Question 3	
		3ai) $LHS = \frac{\sin 2x}{1 - \cos 2x}$ $= \frac{2 \sin x \cos x}{1 - (\cos^2 x - \sin^2 x)}$ $= \frac{2 \sin x \cos x}{1 - \cos^2 x + \sin^2 x}$ $= \frac{2 \sin x \cos x}{\cos^2 x + \sin^2 x - \cos^2 x + \sin^2 x}$ $= \frac{2 \sin x \cos x}{2 \sin^2 x}$ $= \frac{\cos x}{\sin x}$ $= \cot x$	1 mark
	1 mark for both the diagrams		1 mark

2022 Year 11 Mathematics Ext1 AT2 Solutions

Suggested Solution (s)	Comments	Suggested Solution (s)	Comments	
3aii) $\cot 67\frac{1}{2}^\circ = \frac{\sin 2 \times 67\frac{1}{2}^\circ}{1 - \cos 2 \times 67\frac{1}{2}^\circ}$ $\cot 67\frac{1}{2}^\circ = \frac{\sin 135^\circ}{1 - \cos 135^\circ}$ $\cot 67\frac{1}{2}^\circ = \frac{1}{\sqrt{2}} \div \left(1 - \left(-\frac{1}{\sqrt{2}}\right)\right)$ $\cot 67\frac{1}{2}^\circ = \frac{1}{\sqrt{2}} \div \left(\frac{\sqrt{2}+1}{\sqrt{2}}\right)$ $\cot 67\frac{1}{2}^\circ = \frac{1}{\sqrt{2}+1} = \sqrt{2}-1$ 3bi) Area of Sector = $\frac{45^\circ}{360^\circ} \times \pi \times 10^2$ Area of Sector = $\frac{100\pi}{8}$ Area of Sector = $\frac{25\pi}{2}$ bii) Area of shaded portion = $\frac{1}{2} \times 10 \times 10 - \frac{25\pi}{2}$ Area of shaded portion = $\frac{100 - 25\pi}{2}$	1 mark No penalty for not rationalising 1 mark 1 mark	Question 4 i) $T = 24 - Ae^{-kt}$ $\frac{dT}{dt} = -k \times -Ae^{-kt}$ $\frac{dT}{dt} = -k(24 - Ae^{-kt} - 24)$ $\frac{dT}{dt} = -k(T - 24)$ $\therefore T = 24 - Ae^{-kt}$ is a solution ii) when $t = 0, T = -10$ $-10 = 24 - Ae^0$ $\therefore A = 34$ iii) when $t = 90, T = 2^\circ C$ $2 = 24 - 34e^{-k \times 90}$ $\therefore \frac{-22}{-34} = e^{-90k}$ $\log_e \left(\frac{11}{17}\right) = -90k$ $k = -\frac{1}{90} \log_e \left(\frac{11}{17}\right)$ iv) when $t = 180 \text{ min}$ $T = 24 - 34e^{2 \log_e \left(\frac{11}{17}\right)} = 9.76$	1 mark for complete answer 1 mark 1 mark 1 mark 1 mark	

2022 Year 11 Mathematics Ext1 AT2 Solutions

Question 5

$$\begin{aligned} \text{ai) } LHS &= \frac{1}{\sqrt{x}} - \frac{1}{\sqrt{x+h}} \\ &= \frac{\sqrt{x+h} - \sqrt{x}}{\sqrt{x}\sqrt{x+h}} \times \frac{\sqrt{x} + \sqrt{x+h}}{\sqrt{x} + \sqrt{x+h}} \\ &= \frac{x+h-x}{x\sqrt{x+h} + \sqrt{x}(x+h)} \\ &= \frac{h}{\sqrt{x}(x+h) + x\sqrt{x+h}} \end{aligned}$$

$$\begin{aligned} \text{ii) } \frac{d}{dx} \sqrt{\frac{2}{x}} &= \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h} \\ &= \sqrt{2} \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{1}{\sqrt{x+h}} - \frac{1}{\sqrt{x}} \right) \\ &= \sqrt{2} \lim_{h \rightarrow 0} \frac{1}{h} \left(\frac{-h}{\sqrt{x}(x+h) + x\sqrt{x+h}} \right) \\ &= \sqrt{2} \left(\frac{-1}{x\sqrt{x} + x\sqrt{x}} \right) \\ &= \frac{-\sqrt{2}}{2x\sqrt{x}} = \frac{-\sqrt{2x}}{2x^2} \end{aligned}$$

Question 6

$$\begin{aligned} \text{a) } \sin x + \cos x &= -1 \\ \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2} &= -1 \\ \therefore 2t + 1 - t^2 &= -1 - t^2 \\ 2t &= -2 \\ t &= -1 \\ \therefore \frac{x}{2} &= \pi - \frac{\pi}{4} \\ x &= \frac{3\pi}{2} \end{aligned}$$

test: $x = \pi$

$$\sin \pi + \cos \pi = 0 - 1 = -1$$

$$\therefore x = \pi \text{ and } x = \frac{3\pi}{2}$$

$$\text{b) } LHS = \tan 2\sigma (\sin 2\sigma - \tan \sigma)$$

$$\text{let } t = \tan \sigma$$

$$\therefore LHS = \frac{2t}{1-t^2} \left(\frac{2t}{1+t^2} - t \right)$$

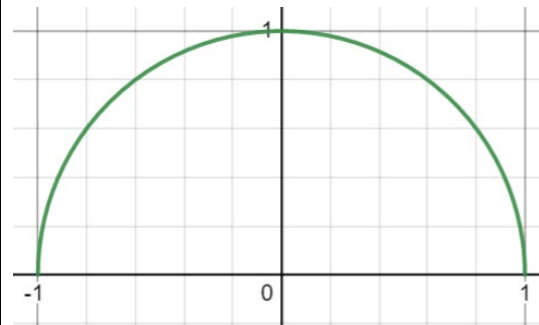
$$= \frac{2t}{1-t^2} \left(\frac{2t-t-t^3}{1+t^2} \right)$$

$$= \frac{2t}{1-t^2} \times \frac{t(1-t^2)}{1+t^2}$$

$$= \frac{2t}{1+t^2} \times t$$

$$= \tan \sigma \sin 2\sigma$$

c)



1 mark for intercepts

1 mark for the correct shape